Integrating Multisubspace Joint Learning With Multilevel Guidance for Cross-Modal Retrieval of Remote Sensing Images

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***Abstract*— In recent years, with the continuous advancement of remote sensing (RS) technology and text processing techniques, there has been a growing abundance of RS images and associated textual data. Combining RS images with their corresponding textual data allows for integrated analysis and retrieval, which holds significant practical implications across multiple application domains, including geographic information systems (GIS), environmental monitoring, and agricultural management. RS images have the characteristics of multitargets and multiscales, and the textual descriptions of these targets are not fully utilized, leading to a decrease in retrieval accuracy. Previous methods have struggled to balance intermodality information interaction and intramodality feature fusion, and they have paid little attention to the consistency of distribution within modalities. In light of this, this article proposes a symmetric multilevel guidance network (SMLGN) for cross-modal retrieval in RS. SMLGN first introduces fusion guidance between local and global within modalities and fine-grained bidirectional guidance between modalities, allowing for the learning of a common semantic space. Furthermore, to address the distribution differences of different modalities within the common semantic space, we design an adversarial joint learning framework and a multiobjective loss function to optimize the SMLGN method and achieve consistency in data distribution. The experimental results demonstrate that the SMLGN method performs well in the task of cross-modal retrieval between RS images and textual data. It effectively integrates the information from both modalities, improving the accuracy and reliability of the retrieval process.**

***Index Terms*— Adversarial learning, feature fusion, modality alignment, multisubspace joint learning, remote sensing (RS) image-text (I2T) retrieval.**

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I. Introduction

With the rapid development of high-spatial-resolution (HSR) satellites and sensors, the amount of remotely sensed image data has increased exponentially. Cross-modal remote sensing (RS) image retrieval has become an emerging direction in the fields of Earth observation and computer vision. Through cross-modal RS image retrieval, more comprehensive and accurate information retrieval and analysis can be achieved. It is widely used in the fields of disaster monitoring [1], land resource utilization [2] and RS image mapping [3].

The main challenge of cross-modal retrieval is the heterogeneity gap between different modalities, i.e., the difference in the distribution of modal data. So how to establish an effective link between different modalities is the main challenge. The key solution for cross-modal retrieval is to find a public space with semantic information in which RS features and text features can not only be directly used to calculate the distance but also provide better retrieval performance. Therefore, designing an efficient method to find a common space for cross-modal retrieval tasks has become the focus of current researchers.

Most previous methods for RS image-text (I2T) retrieval are based on captioning, where text descriptions of RS images are generated, and then these generated texts are compared and matched with query texts [4], [5], [6] to retrieve relevant RS images. The essence of this approach is still a single-modal retrieval and matching task, which cannot directly match different modalities. Additionally, there is a lack of sufficient information interaction between modalities, leading to low retrieval accuracy. Although subsequent researchers have begun to investigate end-to-end retrieval methods, these methods fail to effectively integrate the object information described in the text with the image features, resulting in incomplete alignment of information between modalities.

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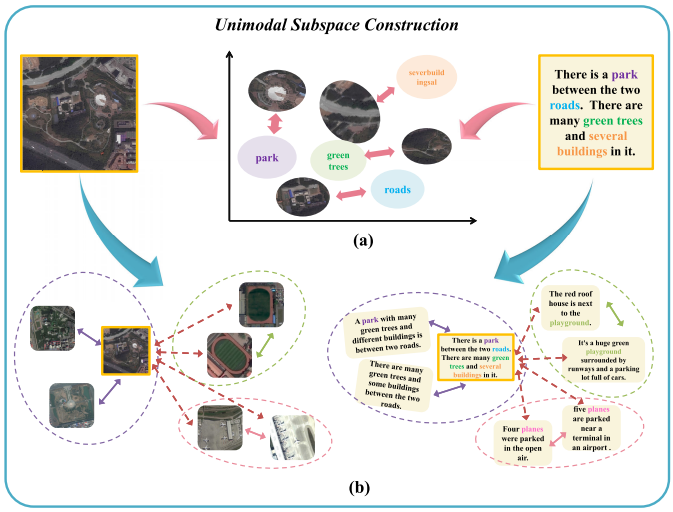


Fig. 1. There are two existing problems in I2T retrieval, namely intramodal discriminability and cross-modal alignment. As shown in (a), arrows represent visually similar image areas and semantically identical words associated with them. As shown in example (b), similar images or text samples are grouped in the same circle. (a) Intermodal alignment. (b) Intramodal discriminability.

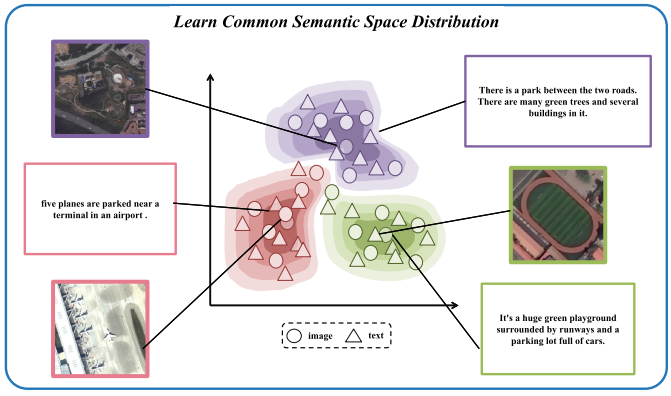


Fig. 2. Data distribution of RS images and text embeddings in the common semantic space. Matching I2T pairs are represented by boxes of the same color between modalities. Similar texts or images cluster together in the same color block regions.

Although there are currently many methods that can achieve cross-modal matching without relying on generated text, they still face the following two problems.

1) While current methods can handle the multiobject and multiscale characteristics of RS images through image feature fusion, they often overlook the local information within the text. If people can pay more attention to the fine-grained alignment between the text and RS images, it would be beneficial for achieving more accurate matching, as shown in Fig. 1(a).
2) Most of the existing research focuses on exploring relationships between different targets within a single-modal space while neglecting the correlation between image-image (I2I) and text-text (T2T) from a global information perspective. Evidently, by minimizing the distance between similar images or texts within a modality, we can achieve better discriminability of different categories within the modality, thus enhancing retrieval accuracy, as illustrated in the diagram, as shown in Fig. 1(b).

In recent years, adversarial learning has excelled in several computer vision tasks. In cross-modal retrieval tasks, adversarial learning can better learn the semantic distributions of different modal data, which in turn helps to eliminate the heterogeneity of different modalities. The multisource cross-modal retrieval network (MCRN) [7] proposed a shared modal transfer module based on modal memories and combined with generative adversarial learning to achieve modality-independent semantic representations. The deep unsupervised cross-modal comparative hashing method (DUCH) [8] first generates unimodal feature representations separately and then learns the joint feature representations between image and text modalities via a hash module. The aforementioned research indicates that adversarial learning provides an effective approach to reducing the discrepancy in heterogeneous data representations. In this regard, we propose a discriminative representation for RS I2T based on adversarial learning to address the challenges of heterogeneity and bridge the gap between different modalities' data distributions.

In this article, we propose a novel symmetric multilevel guidance network (SMLGN) for cross-modal retrieval tasks in RS. Our method utilizes a symmetric network structure for intermodality interaction, allowing the rich semantic information in images to complement the text features, while the text modality helps filter out redundant information in the image features. Additionally, instead of performing multiscale fusion within the image modality, we preserve local feature information and introduce an implicit local mutual guidance unit (ILMGU). Furthermore, to retain global information, we design a locally enhanced global reconstruction cell (LEGRC). Combining the two, we obtain the final multilevel guided joint unit (MLGJU), which enables implicit interaction between modalities at a fine-grained level through the implicit local mutual guidance module. Unlike previous research that employs explicit intermodal local information interaction, our implicit interaction approach allows the model to preserve global information without overly focusing on local details. Finally, we design a multiobjective loss function that integrates adversarial learning and metric learning. This loss function not only optimizes the modality distributions through adversarial learning but also simultaneously constrains intramodal clustering and intermodal interaction. The resulting distribution of image and text embeddings in a common semantic space, as shown in Fig. 2, exhibits evenly distributed modality embeddings. Simultaneously, semantically similar images and texts are clustered together, thereby improving the final retrieval accuracy.

The main contributions of our work are as follows.

1) We propose a novel SMLGN for cross-modal retrieval in RS. We leverage information from local and global subspaces of different modalities to perform modality alignment and guidance both within and between modalities, ultimately obtaining a common space with semantic information. To the best of our knowledge, we are the first to propose a multilevel guidance learning method for cross-modal RS I2T retrieval.
2) Due to the multiscale and multitarget characteristics of RS images and text, we introduce an MLGJU composed of an ILMGU and an LEGRC. This unit allows

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for fine-grained intermodality alignment at the sublevel and implicitly guides the fusion of global information using locally aligned information. This helps prevent the model from losing focus on long-range global information.

3) To align the distributions of different modalities in the common semantic space, we design a multiobjective loss function that combines adversarial learning and metric learning. This loss function not only utilizes adversarial learning to optimize the data distributions but also simultaneously constrains intramodal clustering and intermodal interaction. Experimental results show that the method outperforms current state-of-the-art methods on several mainstream I2T datasets.

II. Related Work

In this section, our focus is on examining previous studies that are most pertinent to our proposed methodology. This includes a review of cross-modal I2T retrieval techniques applied to natural scenes, with subcategories such as deep learning methods utilizing convolutional neural networks (CNNs), Transformers, and generative adversarial networks (GANs). Additionally, we also explore cross-modal I2T retrieval approaches specifically designed for RS scenes.

A. Cross-Modal Image-Text Retrieval

In order to solve the heterogeneous differences between different modal data, most of the research on cross-modal retrieval tends to find a function to map different modal information into a common space. The existing methods include those based on deep neural networks (CNNs), Transformer networks, and GANs.

1) Based-CNNs Methods: In recent research, the emphasis has been placed on leveraging various CNN architectures to effectively capture and model the intricate cross-modal correlations while learning common representations. This is primarily due to the CNNs' robust nonlinear feature extraction capabilities. While deep learning has achieved the representation learning of single modes like text, image, and audio, bridging the gaps between many modes is still a significant difficulty for deep learning techniques. Inspired by hash coding in unimodal image retrieval task [9], [10], hash function can also be introduced in cross-modal image retrieval task. The DVSH model, as described in the paper [11], leveraged the spatial dependencies of images and the temporal dynamics of text sentences. By doing so, it is able to generate compact hash codes for both images and sentences within an end-to-end deep learning framework. This approach facilitated the learning of robust feature representations and cross-modal embeddings, effectively addressing the heterogeneity of different modalities. Carvalho et al. [12] suggested a double-triplet approach to describe both the class-based and retrieval losses simultaneously. This double loss is capable of accounting for both high-level and fine-grained semantic information. A paradigm for fine-tuning based on cooperative retrieval and reranking approaches was proposed by Geigle et al. [13]. All data was encoded using a twin network combined, allowing for quick initial retrieval, and the same data unit was retrieved using a cross-encoder structure. The deep fragment model [14] took into account visual objects, phrase fragments, and information at a finer level. A new fragment-level loss algorithm was also included to go along with the conventional sentence-image ranking loss. Inspired by the aforementioned, compositional cross-modal learning to rank (C2MLR) [15] used maximum interval learning sorting and both local and global alignment to learn multimodal embedded public space from the perspective of pinwise sorting problem optimization. Similarity and diversity induced paired projection (SDPP) [16] extracted the correlation in the public space using the Hilbert–Schmidt independence criterion (HSIC) technique, which removed additional mode-specific information that was irrelevant to the cross-modal retrieval task. Peng et al. [17] built the CMDH, a cross-modal hierarchical network that jointly modeled intermodal and intramodal information using a two-stage learning technique to build shared representations. They also offered the cross-modal correlation learning (CCL) [18] technique, which created a multitask learning strategy for adaptively balancing semantic class constraints within modules and pin-to-module similarity requirements. Furthermore, multigranularity modeling was employed to merge coarse-grained instances with fine-grained patches to enhance the accuracy of cross-modal correlation.

2) Based-Transformer Methods: With the excellent performance of Transformer in NLP tasks, more and more cross-modal tasks are also adopting Transformer for image feature extraction. Previous multimodal pretraining approaches often used CNN structures, such as faster R-CNN, for the visual part. These CNN structures were typically fixed, which limited the potential of the visual-linguistic model. By adopting Transformer structures for both the visual and language modeling parts, the upper limit of the model's performance can be increased. Typically, the contrastive language-image pretraining (CLIP) model [19] utilized the Transformer structure as its fundamental neural network architecture. It leveraged this structure to learn multimodal embedding representations through cotraining text and image data. This approach enabled cross-modal semantic understanding and prediction across various tasks. In addition to this, Dou et al. [20] proposed a multimodal end-to-end TransformER (METER) framework, which focused on how to design and pretrain full Transformer-based visual language models in an end-to-end manner. Following that, Feng et al. [21] proposed a pretrained model called enhanced representation through knowledge integration for visual and language understanding 2 (ERNIE-ViL2), which was trained using multiview learning for text-image (T2I) matching. This model employed a dual-tower architecture where, in each pretraining stage, two visual views and two text views were inputted to their respective encoders. The multiview contrastive learning loss was the sum of four contrastive learning losses: I2I, I2T, T2I, and T2T. Different weights were used to compute the losses for these different views. Guo et al. [22] proposed a method called distilled binary hashing transformer (DBHT) for cross-modal vessel image retrieval. In this model, the visual transformer (ViT) was employed as the feature extractor for target images and was used to generate hash codes. Simultaneously, a unimodal hashing retrieval framework was built, and the hash knowledge was transferred to the other modality. The use of distillation strategies ensured the complete

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transfer of knowledge. Hong and Liu [23] introduced a method called deep semantic preserving visual transformer hashing (DSPVTH), which utilized network structures like ViTs to map different modalities of data into binary hash codes. Simultaneously, it maintained the semantic correlations between different modal data by utilizing the similarity relationships among multiple labels.

3) Based-GANs Methods: The GANs [24] have demonstrated excellent capabilities in modeling data distributions and developing discriminative representations of unimodal data in recent years. Several studies [25], [26], [27], [28] adapted unimodal GANs to simulate multimodal data joint distributions. These methods employed adversarial learning to acquire modal invariant representations by employing a feature generator and a modal discriminator for each modality. Wang et al. [25] presented adversarial cross-modal retrieval (ACMR) after being inspired by game-theoretic principles. The interaction of a feature mapper and a modal classifier lied at the heart of ACMR. The former mapped different modalities into a shared subspace, whereas the latter attempted to discriminate modalities in the subspace and hence controlled the former's learning. The two competed with each other and constantly optimized. In the unsupervised generative adversarial cross-modal hashing (UGACH) [29] model, given the data of one mode, the generative model would try to fit the distribution on the flow structure and select the information data of another mode to challenge the discriminant model. The discriminant model learned to distinguish between the generated data and the real forward data sampled from the correlation graph for better retrieval accuracy. The multistream encoder model [30] maintained the original semantic space's class topology in the encoding feature space while simultaneously having good discriminability in the shared semantic-driven encoding feature space. To put it another way, the model was capable of preserving semantic space while removing modal dependencies. Zero-shot cross-modal retrieval GAN (ZSCRGAN) [31] suggested a new GAN-based zero-snapshot text-to-image (t2i) retrieval model. It used a common space embedding mapper (CSEM) to conduct retrieval without needing additional semantic data from class labels. For two-modal input data, joint feature synthesis and embedding (JFSE) [32] employed two coupled conditional Wassertein GAN modules to create relevant and related multimodal features under the direction of class tag word embedding. By effectively capturing cross-modal correlations, this method not only aided in learning more effective common embedding spaces but also aided in the transfer of knowledge to new classes of multimodal data. The uniqueness of recipe retrieval GAN (R2GAN) [33] was derived from its architectural design, notably from the usage of a GAN with a single generator and dual discriminators, which made the idea of creating an image from a recipe practical. Additionally, a two-level ranking loss in both embedding and image spaces was taken into account depending on the generated images.

B. Cross-Modal Image-Text Retrieval for Remote Sensing Scenes

RS image cross-modal retrieval is developing at a similar rate to natural image cross-modal retrieval. Currently, RS cross-modal retrieval methods may be categorized into three groups based on the various ways of modal interaction [34]: intramodal interaction method, intermodal interaction method, and intramodal and intermodal interaction method. Intramodal inference methods execute self-inference only on the same modality. Self-attention methods are frequently utilized to capture a single mode's semantic representation. In the early studies [35], [36], information from several modalities was immediately encoded and combined to produce a shared semantic representation. Further studies [37], [38] added more information to improve the representation of the modalities. To solve the issue of coarse-grained text description, Mi et al. [37] developed a knowledge-based strategy. In order to combine the multilevel information of images, Yuan et al. [38] presented a network architecture that utilizes both global and local information. To ensure the maximal retention of information in the features, Zhang et al. [39] created a reconstruction module to reconstruct the features after decoupling. Cross-modal interaction methods involve transferring information between multiple modalities, generally via cross-attention processes or shared parameter networks for interactive learning. Lv et al. [40] presented a fusion-based correlation learning model to improve the utilization of cross-modal data correlation. A shared pattern transfer module was used by Yuan et al. [7] to promote communication between various modalities and address semantic heterogeneity between them. The intramodal reasoning and cross-modal interaction methods is a way to integrate the two aforementioned interaction approaches. Cheng et al. [41] suggested a semantic alignment module that effectively examines the hidden correspondences between images and texts. It filtered and optimized data characteristics using attention and gating techniques to provide distinctive feature representations. An asymmetric multimodal feature matching network (AMFMN) was presented by Yuan et al. [42] that not only accepted multiscale feature inputs but also made use of visual features to direct text representation.

III. Proposed Method

A. Problem Definition

Given N paired units $\{I_n, T_n\}_{n=1}^{N}$, where I_n represents the nth RS image instance and T_n represents the nth text instance, where $T_n = \{e_1, e_2, \ldots, e_m\}$, e_i represents the ith word contained in each text instance. The SMLGN method proposed in this article aims to find the common semantic space of RS image modal subspace and text modal subspace while maintaining cross-modal sample similarity. On the one hand, this common space ensures that the features of different modalities can be uniformly distributed in it. On the other hand, it means that similar images and texts are closer to each other. At the same time, the feature embeddings in this common semantic space are directly measurable, i.e., the features therein can be directly used to calculate the degree of similarity using a distance metric function. In particular, the higher the similarity between the nth RS image instance and the nth text instance, the higher their feature embedding similarity score.

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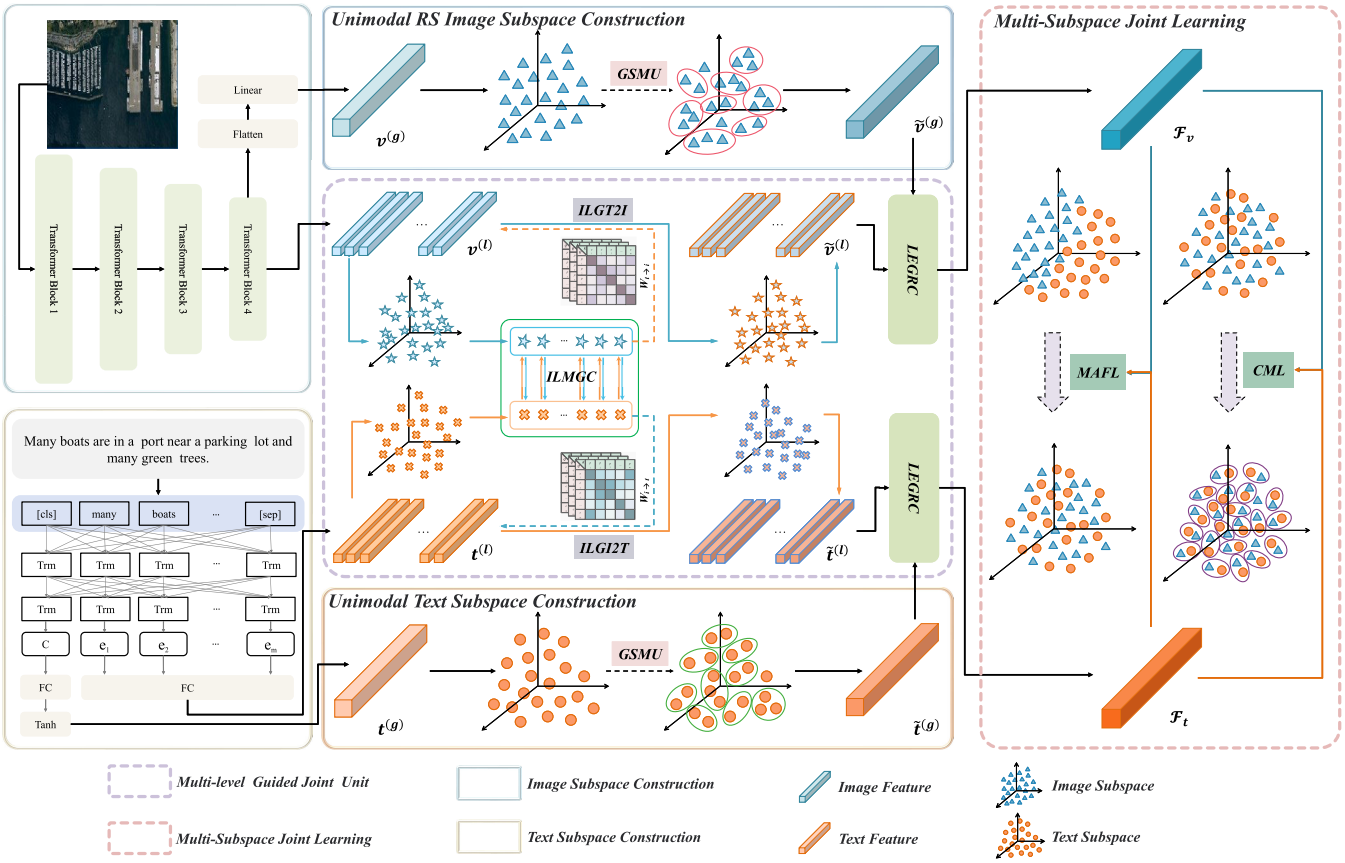


Fig. 3. Overall architecture of an SMLGN. The input of SMLGN is a RS I2T pair. The framework consists of an RS image feature extraction module, a text feature extraction module, a unimodal RS image subspace construction, a unimodal text subspace construction, a multilevel guided joint unit, and a multisubspace joint learning module. This framework can be used for fine-grained alignment and local-global fusion to obtain image features and text features with semantic consistency, modal indistinguishability, and the common semantic space they span.

B. Feature Extraction

Our approach completely uses the Transformer architecture as the backbone network for feature extraction. At the same time, a two-branch structure is used to construct the unimodal original subspace by extracting features from RS images and text, respectively. This is summarized next in terms of both image and text.

1) Image Feature Extraction: For RS images, we utilize a model pretrained by pyramid vision transformer (PVT) [43], [44] on the ImageNet dataset and transfer its parameters to our task. Specifically, the structure of the image subspace construction contains four modules. Each block contains a Transformer encoder layer, and the input image is first divided into a number of patches. The flattened patches are fed into the Transformer layer, and the output is reshaped into a feature mapping as the output of each of these modules. Finally, the output of the fourth Transformer block is spread and fed into the linear projection to obtain the global feature $v^{(g)}$. The process can be represented as

$$\begin{aligned} x^{(1)} &= \text{Transformer}_1(I) \\ x^{(2)} &= \text{Transformer}_2\left(x^{(1)}\right) \\ x^{(3)} &= \text{Transformer}_3\left(x^{(2)}\right) \\ x^{(4)} &= \text{Transformer}_4\left(x^{(3)}\right) \\ v^{(g)} &= \text{Linear}\left(\text{Flatten}\left(x^{(4)}\right)\right) \end{aligned} \tag{1}$$

where $\text{Transformer}_i(\cdot)(i = 1, 2, 3, 4)$ denotes each Transformer block, and $\text{Flatten}(\cdot)$ denotes the spreading operation. $x^{(i)}(i = 1, 2, 3, 4)$ denotes the output of each Transformer block.

RS images differ from natural images, so there is an obvious deficiency in using only the global feature $v^{(g)}$ to represent the picture information. On the one hand, RS images contain a large number of entities. Thus, directly using the global feature $v^{(g)}$ for subsequent retrieval will result in the loss of local information, which makes the retrieval accuracy decrease. On the other hand, RS images have targets at different scales, so single scale features cannot characterize the images well. The PVT method adopts a progressive compression strategy to control the feature map scale by piecewise embedding layers. Inspired by this, we utilize the output of the fourth Transformer block of the PVT model as the local information of the image. Specifically, the local feature $v^{(l)}$ is denoted as

$$v^{(l)} = x^{(4)}. \tag{2}$$

2) Text Feature Extraction: As shown in Fig. 3, the query text T_n is fed into the Bert [45] model as an input vector after performing operations such as split-word preprocessing and inserting two special mask symbols [cls] and [sep]. Trm is an encoder block with a multiheaded attention converter, and the mask symbol [cls] is output as C through the multilayer Trm.

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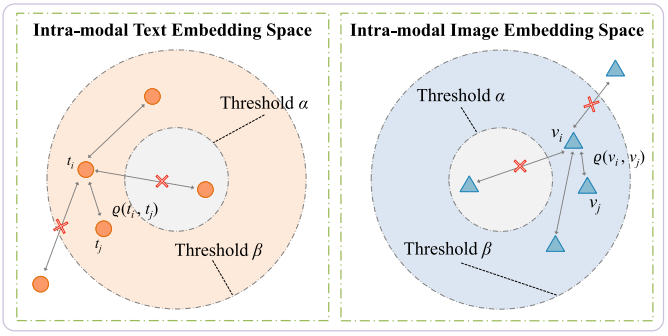


Fig. 4. Intramodal feature embedding space. By calculating the sample distance, the distance between similar RS images or text samples in this space will be narrowed, and the distance between dissimilar samples will be widened. We restrict only the samples within the threshold range by controlling the thresholds α and β.

It then passes through the fully connected layers and activation functions as the global feature $t^{(g)}$ of the text. The process is shown below

$$t^{(g)} = \tanh(\mathrm{FC}(C)) \tag{3}$$

where $\mathrm{FC}(\cdot)$ denotes the fully connected layer and $\tanh(\cdot)$ denotes the activation function.

Each word of the text goes through multiple layers of Trm to get the vector $e = \{e_{k=i}\}_k^m$. It then passes through the fully connected layers to get the local feature $t^{(l)}$ of the text, and the process can be represented as

$$t^{(l)} = \mathrm{FC}(e). \tag{4}$$

C. Unimodal Subspace Construction

The two sets of extracted global and local features separately generate the image and text subspaces, as follows:

$$\begin{aligned} \Omega_{(\mathrm{img})}^{(g)} &= \left\{v_{n=i}^{(g)}\right\}_n^N \qquad \Omega_{(\mathrm{txt})}^{(g)} = \left\{t_{n=i}^{(g)}\right\}_n^N \\ \Omega_{(\mathrm{img})}^{(l)} &= \left\{v_{n=i}^{(l)}\right\}_n^N \qquad \Omega_{(\mathrm{txt})}^{(l)} = \left\{t_{n=i}^{(l)}\right\}_n^N. \end{aligned} \tag{5}$$

In order to fully utilize the original global feature information, we design a global similarity metric unit (GSMU), which is combined with a metric learning approach to construct the intramodal similarity subspace. Taking into account the significant variation within classes and minimal variation between classes in RS images, we expect to increase the interclass variance and decrease the intraclass variance in order to achieve better differentiation of the data. Inspired by Kaya and Bilge [46], we design a contrastive loss to constrain the samples in the unimodal global subspace, as shown in Fig. 4. After GSMU processing, images or texts with a high degree of similarity can be closer together in the global subspace, and images and texts with large differences are farther away. This process helps to find samples of the same class in another modality more accurately when the query modality is entered, greatly improving retrieval accuracy.

We design an intramodal loss function L_{GSMU} in GSMU and then use $L_{\mathrm{GSMU}(I)}$ and $L_{\mathrm{GSMU}(T)}$ to constrain the image modality and text modality, respectively, as defined below

$$L_{\mathrm{GSMU}(I)} = \sum_{i,j\in N} \begin{cases} 0, & \varrho\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) \leq \alpha \\ \varrho\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) & \\ 0, & \beta \leq \varrho\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) \end{cases} \tag{6}$$

$$L_{\mathrm{GSMU}(T)} = \sum_{i,j\in N} \begin{cases} 0, & \varrho\left(t_i^{(g)}, t_{j\neq i}^{(g)}\right) \leq \alpha \\ \varrho\left(t_i^{(g)}, t_{j\neq i}^{(g)}\right) & \\ 0, & \beta \leq \varrho\left(t_i^{(g)}, t_{j\neq i}^{(g)}\right) \end{cases} \tag{7}$$

where $\varrho(\mathrm{x,y})$ is the similarity function between the vectors x and y, and α and β are two thresholds that define the boundaries. The final loss function L_{GSMU} for the GSMU is obtained as

$$L_{\mathrm{GSMU}} = L_{\mathrm{GSMU}(I)} + L_{\mathrm{GSMU}(T)}. \tag{8}$$

Typically, the similarity function ϱ is usually the cosine distance function in retrieval tasks

$$\begin{aligned} \varrho_{\cos}\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) &= 1 - \cos\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) \\ &= 1 - \frac{\sum_{k=1}^{d} v_{i,k}^{(g)} \cdot v_{j\neq i,k}^{(g)}}{\sqrt{\sum {v_{i,k}^{(g)}}^2}\sqrt{\sum {v_{j\neq i,k}^{(g)}}^2}} \end{aligned} \tag{9}$$

where $v_i^{(g)}$ and $v_{j\neq i}^{(g)}$ are the negative pairs of global features of the image modality, and d is the dimension of the global features. The distance metric between global features of the text is also computed as in (9).

The similarity function ϱ can be substituted with other similarity measures, such as the Manhattan distance ($L1$), Euclidean distance ($L2$), and mean squared displacement (MSD) function. Their calculation formulas are as follows.

1) *Manhattan Distance (L1):*

$$\begin{aligned} \varrho_{L1}\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) &= L1\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) \\ &= \sum_{k=1}^{d} \left| v_{i,k}^{(g)} - v_{j\neq i,k}^{(g)} \right|. \end{aligned} \tag{10}$$

2) *Euclidean Distance (L2):*

$$\begin{aligned} \varrho_{L2}\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) &= L2\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) \\ &= \sum_{k=1}^{d} \sqrt{\left(v_{i,k}^{(g)} - v_{j\neq i,k}^{(g)}\right)^2}. \end{aligned} \tag{11}$$

3) *MSD:*

$$\begin{aligned} \varrho_{\mathrm{MSD}}\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) &= \mathrm{MSD}\left(v_i^{(g)}, v_{j\neq i}^{(g)}\right) \\ &= \sum_{k=1}^{d} \left(v_{i,k}^{(g)} - v_{j\neq i,k}^{(g)}\right)^2. \end{aligned} \tag{12}$$

The impact of these different similarity functions on retrieval accuracy will be demonstrated in Section IV-F2.

The image and text global features in $\Omega_{(\mathrm{img})}^{(g)}$ and $\Omega_{(\mathrm{txt})}^{(g)}$ are sequentially input into the GSMU to be constrained by using $\mathcal{L}_{\mathrm{MS}}$. After this process, the constrained unimodal global features are noted as $\widetilde{v}^{(g)}$ and $\widetilde{t}^{(g)}$, respectively. They are tensored into reconstructed global subspaces, denoted, respectively, as $\Omega_{(\widetilde{\mathrm{img}})}^{(g)}$ and $\Omega_{(\widetilde{\mathrm{txt}})}^{(g)}$.

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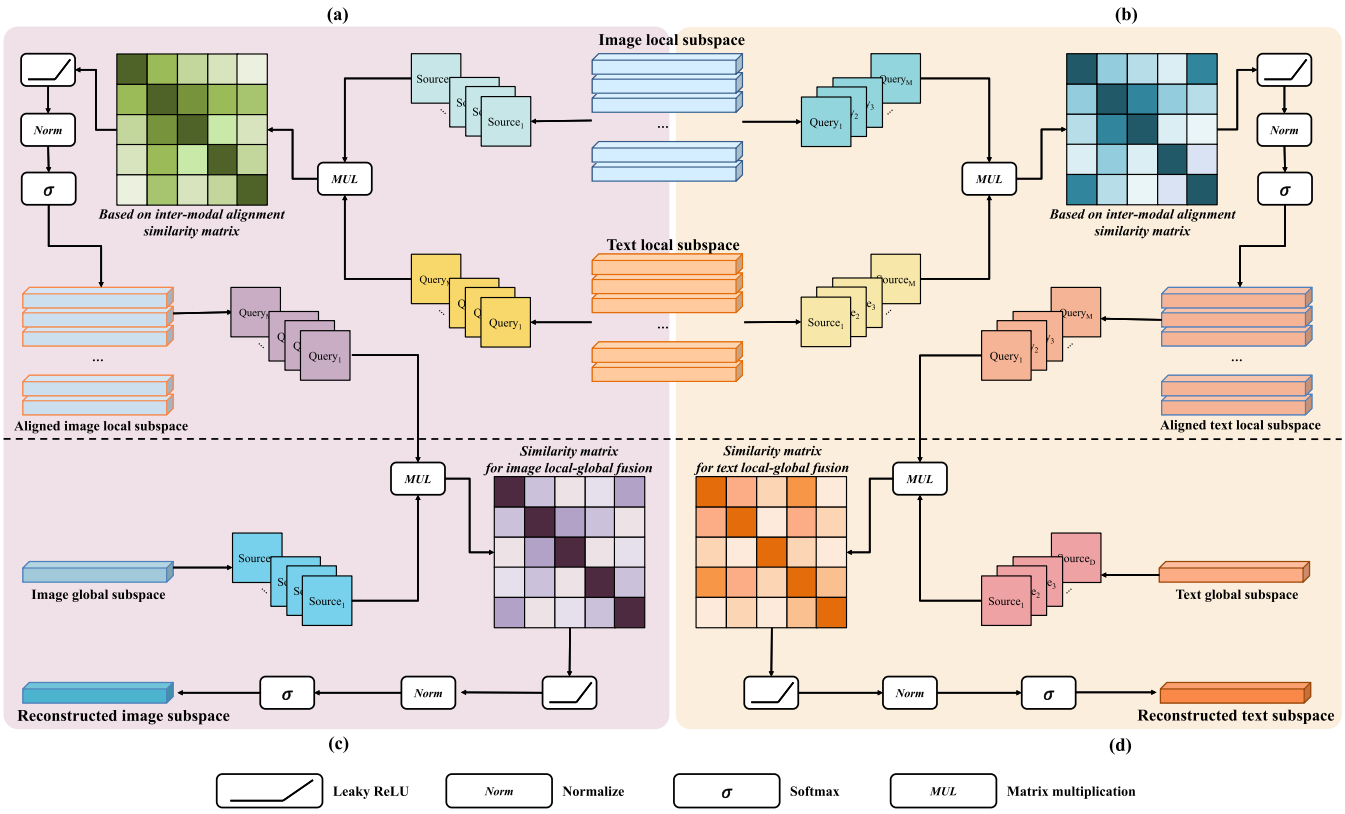


Fig. 5. Overall architecture of the multilevel guidance unit. (a) and (b) ILMGU. (c) and (d) Locally enhanced global reconstruction cell (LEGRC). The output features of this module will be the input of multisubspace joint learning. (a) Text to image implicit alignment guidance. (b) Image to text implicit alignment guidance. (c) Image locally enhanced global reconstruction. (d) Text locally enhanced global reconstruction.

D. Multilevel Guided Joint Unit

In order to fully utilize the rich scale information in RS images and the descriptive information of local details in the text, we have designed an MLGJU. This unit consists of the ILMGU and the LEGRC. The ILMGC allows for fine-grained alignment between local RS images and text by implicitly calculating the similarity between them from the same semantic center. It not only achieves alignment at a fine level but also guides the global information using locally aligned information. The LEGRC incorporates guided local information into global information, generating modality features that consider both global and fine-grained details. This approach effectively addresses the issues of inadequate alignment and detail loss in current cross-modal retrieval tasks. Next, we will elaborate on each of the two guidance units.

1) Implicit Local Mutual Guidance Cell: While most of the existing methods are realized by explicitly aligning local features between image and text, our proposed ILMGU can provide implicit bidirectional guided alignment between modalities. Semantic information is transferred between modalities through ILMGC to compensate for the distributional differences of modal data, as illustrated in Fig. 5(a) and (b). Given a set of image and text local feature pairs $\{v^{(l)},\ t^{(l)}\}$, where $v^{(l)} = \{p_1, p_2, \ldots, p_m\}$, m denotes the number of patches per image, $t^{(l)} = \{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_k\}$, k denotes the number of words each text contains.

a) Implicit local guidance for text to image: We design a module called implicit local guidance for text to image (ILGT2I) for word features in text to guide the alignment of image patches information in images to fully utilize the local attributes of the query text to locate relevant or important regions in images. The realization process is shown in Fig. 5(a), and the specific description is unfolded as follows.

First, each set of image local features in the subspace $\Omega^{(l)}_{(\mathrm{img})}$ is used as *Source* keys. It needs to be guided by words related to the local information of the image, thereby complementing the textual semantics in the image subspace. Then each set of text local features in the subspace $\Omega^{(l)}_{(\mathrm{txt})}$ is used as *Query* keys. It will find the local position of the image associated with the corresponding word to improve the retrieval effectiveness of the local information of that image to strengthen the connection between modalities.

The cosine similarity function is employed to compute the similarity s_{ab} between the vectors in these two local subspaces. This step is calculated as

$$s_{ab} = \frac{p_a \cdot \tau(\varepsilon_b)}{\|p_a\|\|\varepsilon_b\|} \tag{13}$$

where s_{ab} denotes the similarity between the ath region and the bth word, τ represents the transpose function, and $\|\cdot\|$ denotes the length of the vector.

The guidance matrix $S_{t\to i}$ for a given I2T pair consists of all the similarities between regions and words. This step is

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calculated as follows:

$$S_{t\to i}=\begin{bmatrix} s_{11} & \cdots & s_{1k} \\ \vdots & \ddots & \vdots \\ s_{m1} & \cdots & s_{mk} \end{bmatrix}. \tag{14}$$

This guidance matrix is activated by the Leaky rectified linear unit (ReLU) function, and then it is normalized. Finally, it is then operated by the activation function to obtain the weight matrix $W_{t\to i}$. The above process can be formulated as

$$W_{t\to i}=\text{softmax}(\text{Norm}(\text{ReLU}(S_{t\to i}))) \tag{15}$$

where $\text{ReLU}(\cdot)$ denotes the Leaky ReLU function, $\text{Norm}(\cdot)$ denotes $L2$ normalization, and $\text{softmax}(\cdot)$ denotes the activation function.

We multiply the feature $v^{(l)}$ in the local subspace of the image and this weight matrix $W_{t\to i}$ to finally obtain the reconstructed local features $\widetilde{v}^{(l)}$. The specific process is demonstrated as follows:

$$\widetilde{v}^{(l)}=W_{t\to i}\cdot\tau\left(v^{(l)}\right). \tag{16}$$

b) Implicit local guidance for image to text: Compared with text features, RS images are richer in semantic information. So only relying on text to guide images is not able to fully utilize the information in the local subspace of RS images. Next, we design a module called ILGI2T to utilize the rich semantic information of images. The realization process is illustrated in Fig. 5(b).

Similar to above, the local features of the text in the subspace $\Omega^{(l)}_{(\text{txt})}$ are used as *Source* keys. It needs to be guided by the local semantic information of the image, thus introducing richer information of the image in the text. Next, the local features of each set of images in the subspace $\Omega^{(l)}_{(\text{img})}$ serve as the *Query* keys. It needs to find the word most relevant to the local information of each image to strengthen the text's focus on the local information of the image.

Likewise, we compute the cosine similarity $\hat{s}_{cd}$ between vectors in two local subspaces. The formula for this step is written as

$$\hat{s}_{cd}=\frac{\varepsilon_c\cdot\tau(p_d)}{\|\varepsilon_c\|\,\|p_d\|} \tag{17}$$

where $\hat{s}_{cd}$ denotes the similarity between the cth word and the dth region.

Again, construct the guidance matrix $S_{i\to t}$ for a given T2I pair and the weight matrix $W_{i\to t}$ for which the image guides the text. This process is computed as follows:

$$S_{i\to t}=\begin{bmatrix} \hat{s}_{11} & \cdots & \hat{s}_{1m} \\ \vdots & \ddots & \vdots \\ \hat{s}_{k1} & \cdots & \hat{s}_{km} \end{bmatrix} \tag{18}$$

$$W_{i\to t}=\text{softmax}(\text{Norm}(\text{ReLU}(S_{i\to t}))). \tag{19}$$

In order to get the textual reconstructed local features $\widetilde{t}^{(l)}$, we multiple the word features $t^{(l)}$ with this weight matrix $W_{i\to t}$. The procedure is shown below

$$\widetilde{t}^{(l)}=W_{i\to t}\cdot\tau\left(t^{(l)}\right). \tag{20}$$

All the reconstructed local features $\widetilde{v}^{(l)}$ and $\widetilde{t}^{(l)}$, respectively, generate a new reconstructed visual local subspace $\Omega^{(l)}_{(\widetilde{\text{img}})}$ and a new textual local subspace $\Omega^{(l)}_{(\widetilde{\text{txt}})}$. After the ILMGC, the reconstructed local subspaces are able to contain information of another modality. This step is noted as

$$\begin{aligned}\Omega^{(l)}_{(\widetilde{\text{img}})}&=\left\{\widetilde{v}^{(l)}_{n=i}\right\}^N_n\\ \Omega^{(l)}_{(\widetilde{\text{txt}})}&=\left\{\widetilde{t}^{(l)}_{n=i}\right\}^N_n.\end{aligned} \tag{21}$$

2) Global Subspace Construction With Local Guidance: Previously, in retrieval models that use both global and local features, some approaches used a two-branch structure to directly output local and global features to calculate the similarity separately. This approach fails to unify global and local features and also generates redundant computations that reduce the retrieval efficiency. Therefore, we propose an LEGRC. It can not only utilize the important semantic information in the local subspace to supplement the missing local information in the global subspace but also introduce intermodal guidance information. The specific process is shown in Fig. 5(c) and (d).

The reconstructed image global feature $\widetilde{v}^{(l)}$ and the reconstructed text local feature $\widetilde{t}^{(l)}$ separately serve as *Query* keys. They need to help the corresponding global subspace complement the intermodal local alignment information. Meanwhile, the reconstructed image global feature $\widetilde{v}^{(g)}$ and the reconstructed text global feature $\widetilde{t}^{(g)}$ are, respectively, used as *Source* keys. The cosine similarity function is utilized to compute the similarity between embeddings in the global and local subspaces. This process can be formulated as

$$\begin{aligned}s^{(\text{gl})}_v&=\frac{\widetilde{v}^{(g)}\cdot\tau\left(\widetilde{v}^{(l)}\right)}{\left\|\widetilde{v}^{(g)}\right\|\left\|\widetilde{v}^{(l)}\right\|}\\ s^{(\text{gl})}_t&=\frac{\widetilde{t}^{(g)}\cdot\tau\left(\widetilde{t}^{(l)}\right)}{\left\|\widetilde{t}^{(g)}\right\|\left\|\widetilde{t}^{(l)}\right\|}\end{aligned} \tag{22}$$

where $s^{(\text{gl})}_v$ and $s^{(\text{gl})}_t$ denote the similarity between all local information and global features in an image or a text. The weight matrix containing the local information can be obtained by doing the following for $s^{(\text{gl})}_v$ and $s^{(\text{gl})}_t$

$$\begin{aligned}\mathcal{W}_v&=\text{softmax}\left(\text{Norm}\left(\text{ReLU}\left(s^{(\text{gl})}_v\right)\right)\right)\\ \mathcal{W}_t&=\text{softmax}\left(\text{Norm}\left(\text{ReLU}\left(s^{(\text{gl})}_t\right)\right)\right).\end{aligned} \tag{23}$$

The obtained weight matrices $\mathcal{W}_v$ and $\mathcal{W}_t$ are multiplied with the reconstructed global features $\widetilde{v}^{(g)}$ and $\widetilde{t}^{(g)}$. Finally, the image feature $\mathcal{F}_v$ and text feature $\mathcal{F}_t$ that take into account the local-global information and intermodal alignment information are obtained. The process is as follows:

$$\begin{aligned}\mathcal{F}_v&=\mathcal{W}_v\cdot\tau\left(v^{(g)}\right)\\ \mathcal{F}_t&=\mathcal{W}_t\cdot\tau\left(t^{(g)}\right).\end{aligned} \tag{24}$$

The final space that is consisted of all the image features $\mathcal{F}_v$ and text features $\mathcal{F}_t$ output by LEGRC is the final unimodal image subspace $\Omega_{\mathcal{F}_v}$ and text subspace $\Omega_{\mathcal{F}_t}$ which we need to perform joint learning. This step is formulated as follows:

$$\begin{aligned}\Omega_{\mathcal{F}_v}&=\{\mathcal{F}_{vn=i}\}^N_n\\ \Omega_{\mathcal{F}_t}&=\{\mathcal{F}_{tn=i}\}^N_n.\end{aligned} \tag{25}$$

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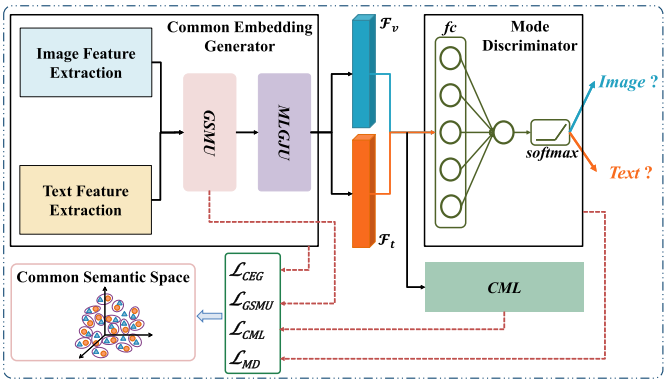


Fig. 6. Architecture of multisubspace joint learning. The features $\mathcal{F}_v$, $\mathcal{F}_t$ in the image subspace $\Omega_{\mathcal{F}_v}$ and the text subspace $\Omega_{\mathcal{F}_t}$ are processed by adversarial learning and modal matching. Then multiobjective loss is used for joint optimization to obtain the final common semantic space.

E. Multisubspace Joint Learning

In general, an ideal cross-modal shared representation should possess two important characteristics that are crucial for cross-modal retrieval.

1) It should demonstrate semantic interoperability, indicating its capacity to maintain semantic consistency across different modalities.
2) It should be modalities-agnostic, effectively addressing the heterogeneity divide by not discriminating between different modes.

To fulfill these characteristics, we design a multisubspace joint learning unit to search for an appropriate shared semantic space. We employ two loss branches to drive the network training for the individual modality image subspace $\Omega_{\mathcal{F}_v}$ and the individual modality text subspace $\Omega_{\mathcal{F}_t}$. These branches are known as cross-modal matching learning (CML) and modality adversarial fusion learning (MAFL), respectively. Its structure is shown in Fig. 6.

1) Cross-Modal Matching Learning: In order to maintain similarity across modal samples, we define the CML inspired by the previous study [47]. It keeps similar RS I2T pairs as close together as possible in the common semantic space, and dissimilar I2T pairs as far apart as possible. The loss function for CML can be expressed as

$$\mathcal{L}_{\mathrm{CML}} = \sum_{\widehat{\mathcal{F}_t}} \left[\kappa - S(\mathcal{F}_v, \mathcal{F}_t) + S\left(\mathcal{F}_v, \widehat{\mathcal{F}_t}\right)\right]_+ + \sum_{\widehat{\mathcal{F}_v}} \left[\kappa - S(\mathcal{F}_v, \mathcal{F}_t) + S\left(\widehat{\mathcal{F}_v}, \mathcal{F}_t\right)\right]_+ \quad (26)$$

where κ is the marginal threshold parameter. The first sum considers all negative text features $\widehat{\mathcal{F}_t}$ of the image feature $\mathcal{F}_v$, the second sum considers all negative image features $\widehat{\mathcal{F}_v}$ of the sentence $\mathcal{F}_t$, and $[x]_+ = \max(x, 0)$.

2) Modality Adversarial Fusion Learning: This branch can be seen as a modality adversarial network, while the other parts of the SMLGN network before multisubspace joint learning in Fig. 3 constitute a common embedding generator (CEG). We have designed the following loss function to constrain the CEG:

$$\mathcal{L}_{\mathrm{CEG_}v} = -\frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{2}\log(\mathrm{Dis}((\mathcal{F}_v)_i)) + \frac{1}{2}\log(1-\mathrm{Dis}((\mathcal{F}_v)_i))\right]$$

$$\mathcal{L}_{\mathrm{CEG_}t} = -\frac{1}{N}\sum_{j=1}^{N}\left[\frac{1}{2}\log\left(\mathrm{Dis}\left((\mathcal{F}_t)_j\right)\right) + \frac{1}{2}\log\left(1-\mathrm{Dis}\left((\mathcal{F}_t)_j\right)\right)\right]$$

$$\mathcal{L}_{\mathrm{CEG}} = \frac{1}{2}\mathcal{L}_{\mathrm{CEG_}v} + \frac{1}{2}\mathcal{L}_{\mathrm{CEG_}t} \quad (27)$$

where $\mathrm{Dis}(\cdot)$ represents the modality discriminator network.

The purpose of the modality discriminator is to differentiate between different modalities, while the CEG aims to confuse the modality discriminator by reducing the cross-modal representation differences. We design a fully connected network as the modality discriminator, with the image features $\mathcal{F}_v$ and text features $\mathcal{F}_t$ generated by the CEG as inputs. We calculate the modality discrimination loss using the following formula:

$$\mathcal{L}_{\mathrm{MD_}v} = \frac{1}{N}\sum_{i=1}^{N}\left[m_i \cdot \log(\mathrm{Dis}((\mathcal{F}_v)_i)) + (1-m_i)\cdot\log(1-\mathrm{Dis}((\mathcal{F}_v)_i))\right]$$

$$\mathcal{L}_{\mathrm{MD_}t} = \frac{1}{N}\sum_{j=1}^{N}\left[m_j \cdot \log\left(\mathrm{Dis}\left((\mathcal{F}_t)_j\right)\right) + \left(1-m_j\right)\cdot\log\left(1-\mathrm{Dis}\left((\mathcal{F}_t)_j\right)\right)\right]$$

$$\mathcal{L}_{\mathrm{MD}} = \frac{1}{2}\mathcal{L}_{\mathrm{MD_}v} + \frac{1}{2}\mathcal{L}_{\mathrm{MD_}t} \quad (28)$$

where $m_{i/j} = \{0,\ 1\}$, 0 represents the image modality and 1 represents the text modality.

The final optimization objective of our joint multisubspace learning module is as follows:

$$\min_{\theta \notin \mathrm{MD}} (\mathcal{L}_{\mathrm{CML}} + \mu \cdot \mathcal{L}_{\mathrm{CEG}} + \nu \cdot \mathcal{L}_{\mathrm{GSMU}})$$
$$\max_{\theta \in \mathrm{MD}} (\mathcal{L}_{\mathrm{MD}}) \quad (29)$$

where μ and ν are the hyperparameters in the optimization process, and the above process parameters θ are optimized by Adam. The space tensored by the optimized features is the public semantic space we expect.

IV. Experiments and Discussion

This section thoroughly assesses the suggested SMLGN. The built dataset is introduced first, and then the evaluation metrics and implementation details are provided. Next, comparison and ablation tests are carried out to qualitatively and quantitatively show the validity of SMLGN.

A. Datasets and Evaluation Metrics

In this article, three public datasets, namely the UCM dataset, the RSICD dataset, and the RSITMD dataset, are selected to validate the validity of the model. The UCM dataset is one of the earliest proposed datasets for the retrieval of remotely sensed images. Its images are from the USGS National Map, in which there are 2100 images of RS feature types, containing 21 scenes, and each sheet is

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256 × 256 pixels. The RSICD dataset is an extensive and diverse collection of RS images for image captioning tasks. It consists of 10 921 images and 30 different scenes, making it the most comprehensive dataset available in this field. The RSITMD dataset is a fine-grained dataset that has been optimized for RS cross-modal I2T retrieval. A total of 4743 pictures, 23 715 captions, and 24 scenarios are included in this dataset, some of which were chosen from the RSICD dataset and others from Google Earth.

The recall rate at K ($R@K$) is used to evaluate model performance for the image retrieval of texts and the text retrieval of images. The better the performance, the higher the value of $R@K$. The average of numerous $R@k$, which intuitively indicates the overall performance of the technique, was then represented using mean rank (mR).

B. Experiment Details

All experiments are performed on MindSpore and PyTorch [48] running with an NVIDIA GeForce RTX 3090, Intel[1] Xeon[1] Gold 6133 CPU@2.50 GHz × 80 CPU and 125-GB RAM. The input RS image size is changed to 256 × 256 through a series of data preprocessing operations. For images, the global embedding and local embedding dimensions of images are both 256. For text, both sentence embedding and word embedding dimensions of text are also 256. In this way, the dimensions of RS images and query text can be consistent in subsequent feature interactions. Regarding the hyperparameters in SMLGM, we use fivefold cross validation in training and set $\mu = 30$ and $\nu = 70$ by default, and the influence of them is analyzed in Section IV-E. For the rest of the hyperparameters, we set $\kappa = 0.2$ as in [47], and $\alpha = 0.05$, $\beta = 0.5$ by default. Adam is chosen as the optimizer to train the network for up to 30 epochs with a batch size of 32. During the MAFL training, the learning rate decays from $1e^{-6}$ and decreases by 0.5 every 5 epochs. The learning rate for the remaining network parameters decays from $1e^{-5}$ and decreases by 0.5 every 5 epochs. To train the SMLGN network, Adam is selected as the optimizer for a maximum of 30 epochs, using a batch size of 32. For the MAFL training, the learning rate starts at $1e^{-6}$ and decreases by 0.5 every 5 epochs. As for the remaining network parameters, their learning rate starts at $1e^{-5}$ and also decreases by 0.5 every 5 epochs.

C. Baselines

We have selected several cutting-edge models for I2T matching and cross-modal retrieval models for RS images as comparative benchmarks to validate the effectiveness of our models, as follows.

1) *Visual-semantic embeddings (VSE++) [49]:* VSE++ uses VGG19 and bidirectional gated recurrent unit (GRU) to extract image and text features, respectively. And then utilize triple loss to train I2T matching model.
2) *Stacked cross attention network (SCAN) [50]:* SCAN employs a bottom-up approach to extract local features by incorporating salient target detection. Performance evaluation includes SCAN t2i and SCAN image-to-text (i2t) methods compared with other approaches.
3) *Cross-modal adaptive message passing (CAMP) [51]:* CAMP introduces an adaptive information transfer mechanism to find key points in images and texts while suppressing unmatched information. In this experiment, we compare the CAMP-bce method developed in this article with the CAMP-triplet method.
4) *Multi-modal tensor fusion network (MTFN) [52]:* MTFN trains a precise similarity function by directly incorporating I2T fusion and T2T fusion. During the testing phase, MTFN utilizes a generalized cross-modal reordering scheme to enhance the efficiency of the model.
5) *Lightweight multiscale cross-modal retrieval (LW-MCR) [53]:* LW-MCR proposes a new lightweight multiscale retrieval algorithm using a hidden supervised optimization method based on knowledge distillation by acquiring dark knowledge from the teacher network's multilevel layers and representation layers.
6) *AMFMN [42]:* AMFMN combines multiscale fusion and redundant information filtering structures to extract high-level image features. And the image features are used to guide the text features to realize modal interaction.
7) *CLIP [19]:* CLIP is a pretraining method based on contrasting T2I pairs. It employs a single-stream model that uses a single structure to obtain both image features and text features after modal interaction.

D. Comparisons

Our proposed SMLGN method shows significant improvements in the precision of cross-modal retrieval on the RSITMD I2T dataset compared to other state-of-the-art techniques, as demonstrated in Table I on the left side. As evident from Table I, first, it can be observed that the performance of general-domain retrieval methods falls slightly behind that of RS models, with a slight gap in accuracy. Considering the emergence of multimodal pretraining models, we compare our approach with CLIP. It can be seen that the performance of multimodal pretraining models reaches its optimum on multiple metrics. Comparing these three models with ours, Table I reveals that our SMLGN model achieves an accuracy improvement of 9.04% and 7.9% on the mR metric over LM-MCR and AMFMN, respectively. Additionally, SMLGN outperforms the best CLIP model by 6.95%. The outcomes obtained highlight the considerable benefits of our approach in comparison to other leading-edge techniques.

The accuracy results of our SMLGN method and other state-of-the-art techniques for cross-modal retrieval on the RSICD RS I2T dataset are shown in Table I on the right side. In both the I→T and T→I tasks, the SMLGN method has achieved significant improvements over the current state-of-the-art models in various metrics. For example, compared to general-domain cross-modal retrieval models, SMLGN has increased the average precision of R@1 (I→T) from 4.66% to 8.87%. For the LW-MCR and AMFMN models designed based on remotely sensed data, the SMLGN has improved the

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TABLE I
EXPERIMENTAL RESULTS BETWEEN SMLGN AND COMPARED STATE-OF-THE-ARTS ON THE RSITMD AND RDICD I2T DATASETS

Method		RSITMD dataset							RSICD dataset						
		I→T			T→I			mR	I→T			T→I			mR
		R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
General retrieval methods	VSE++	10.38	27.65	39.60	7.79	24.87	38.67	24.83	3.38	9.51	17.46	2.82	11.32	18.10	10.43
	SCAN t2i	10.18	28.53	38.49	10.10	28.98	43.53	26.64	4.39	10.90	17.64	3.91	16.20	26.49	13.25
	SCAN i2t	11.06	25.88	39.38	9.82	29.38	42.12	26.82	5.85	12.89	19.84	3.71	16.40	26.73	14.23
	CAMP-triple	11.73	26.99	38.05	8.27	27.79	44.34	26.20	5.12	12.89	21.12	4.15	15.23	27.81	14.39
	CAMP-bce	9.07	23.01	33.19	5.22	23.32	38.36	22.03	4.20	10.24	15.45	2.72	12.76	22.89	11.38
	MTFN	10.40	27.65	36.28	9.96	31.37	45.84	26.92	5.02	12.52	19.74	4.90	17.17	29.49	14.81
Remote sensing retrieval methods	LW-MCR	9.73	26.77	37.61	9.25	34.07	54.03	28.58	4.39	13.35	20.29	4.30	18.85	32.34	15.59
	AMFMN	10.63	24.78	41.81	11.51	34.09	54.87	29.72	5.39	15.08	23.40	4.90	18.28	31.44	16.42
Multi-modal pre-training methods	CLIP	14.89	33.55	45.64	9.81	32.68	47.43	30.67	7.96	21.41	32.48	5.42	20.02	33.19	20.08
	SMLGN	**17.26**	**39.38**	**51.55**	**13.19**	**43.94**	**60.40**	**37.62**	**8.87**	**25.53**	**37.24**	**7.85**	**27.14**	**42.58**	**24.87**

TABLE II
EXPERIMENTAL RESULTS BETWEEN SMLGN AND COMPARED STATE-OF-THE-ARTS ON THE UCM I2T DATASET

Method	UCM dataset						
	I→T			T→I			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
VSE++	12.38	44.76	65.71	10.10	31.80	56.85	36.93
SCAN t2i	14.29	45.71	67.62	12.76	50.38	77.24	44.67
SCAN i2t	12.85	47.14	69.52	12.48	46.86	71.71	43.43
CAMP-triple	10.95	44.29	65.71	9.90	46.19	76.29	42.22
CAMP-bce	14.76	46.19	67.62	11.71	47.24	76.00	43.92
MTFN	10.47	47.62	64.29	14.19	52.38	78.95	44.65
LW-MCR	**18.10**	47.14	63.81	13.14	50.38	79.52	45.35
AMFMN	16.67	45.71	68.57	12.86	**53.24**	79.43	46.08
CLIP	13.81	47.14	62.38	9.90	46.19	73.14	42.10
SMLGN	12.86	**49.52**	**75.71**	**14.29**	52.76	**84.67**	**48.30**



Fig. 7. Several cases of query text retrieval of RS images. It shows the top five most relevant retrieved images, with ground truth in the red box.

accuracy of R@1 (I→T) by 4.48% and 3.48%, respectively. In addition, compared to the current best-performing model CLIP, SMLGN also achieves better performance, with the mR metric increasing from 20.08% to 24.87%. As one of the large-scale publicly available RS multimodal I2T datasets, RSICD showcases the strong performance of our SMLGN method, further validating the effectiveness of our model.

Table II presents the scores achieved by each model on the UCM dataset. Due to the differences in data distribution in the UCM dataset, the performance of RS retrieval models is superior to traditional retrieval models and multimodal pretraining models. Our method achieves the best results in terms of the mR metric and several other metrics except metric R@1 (I→T). Specifically, the mR metric is 2.24% higher than that of the best-performing AMFMN model among other methods. Moreover, the UCM dataset has a smaller scale compared to the other two datasets, indicating that our method exhibits good performance and robustness on small datasets as well.

In order to visually showcase the retrieval results of cross-modal retrieval and demonstrate the performance of our proposed method from a qualitative perspective, Figs. 7 and 8 present the results of our approach in the form of qualitative experiments. It can be observed that SMLGN can effectively retrieve semantically related instances of the other modality, with the majority of the top five retrieved instances being correct. These results further highlight the superiority of the proposed SMLGN method.

In addition, we also visualized the search results with other recent methods, as shown in Figs. 9 and 10. When using text to retrieve images, it can be seen from Fig. 9 that our method can return the ground truth image in the first place, and the remaining pictures and text descriptions are as semantically consistent as possible. At the same time, the retrieval results of VSE++, SCAN, CAMP and MTFN in the general field are worse than those in the RS field. Not only are the ground truth images returned in a lower order, but they also retrieve images that are less semantically related. LW-MCR and AMFMN search results also include irrelevant images. CLIP is better than other methods in retrieval tasks by pretraining a large amount of data, but our method is more suitable for RS retrieval tasks. As shown in Fig. 10, in the task of image text retrieval, our method can accurately find the five texts corresponding to the image. There are many weakly relevant texts in the search results of other methods. That is, our

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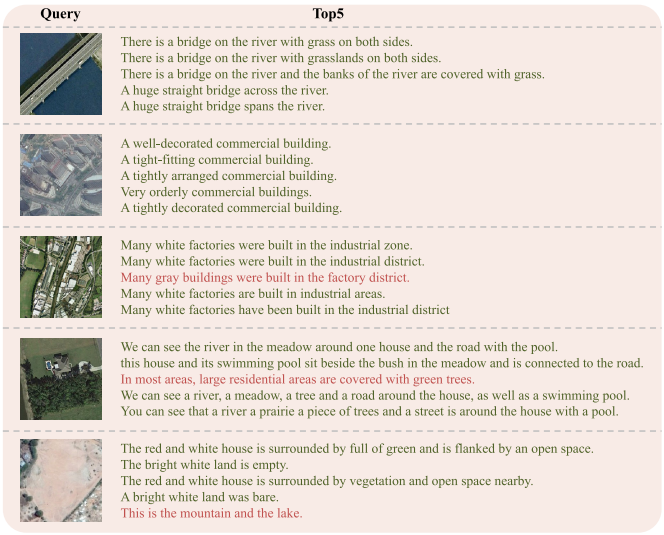


Fig. 8. Several cases of query image retrieval texts. It shows RS images as an example of queries matching relevant text captions, and returns the top five results. The results in green font represent correct matches, and the red font represents incorrect search results.

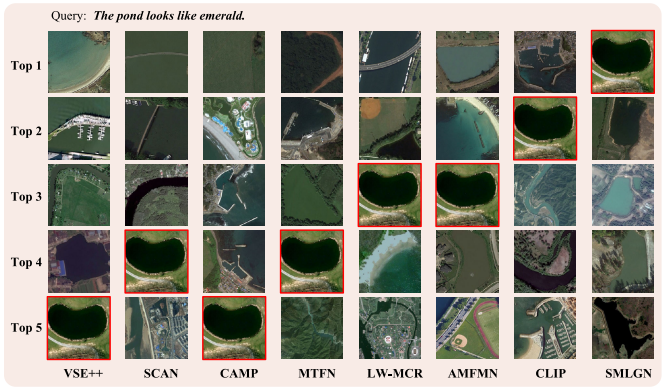


Fig. 9. Comparison results of all methods in the text retrieval images task. It shows the top five most relevant retrieved images, with ground truth in the red box.

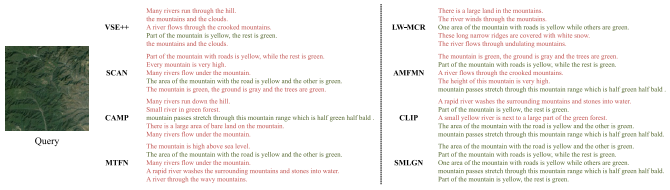


Fig. 10. Comparison results of all methods in the image retrieval texts task. It shows RS images as an example of queries matching relevant text titles and returns the top five results. The results in green font represent correct matches, and the red font represents incorrect search results.

approach is able to focus more comprehensively on complex text, thereby establishing fine-grained alignment with images, and ultimately returning a more appropriate description of the text.

Based on the experiments mentioned above, we believe the effectiveness of SMLGN can be attributed to the following factors.

1) The proposed GSMU performs discriminative clustering within each modality, helping maintain the similarity of data within the same modality in the final shared subspace.
2) The proposed MLGJU aligns and fuses information at both local and global levels. This framework allows for the utilization of fine-grained information between modalities while considering both fine-grained and holistic aspects.
3) The proposed MAFL introduces the concept of adversarial learning, enabling different modalities to be uniformly distributed in the shared space and share semantic information.

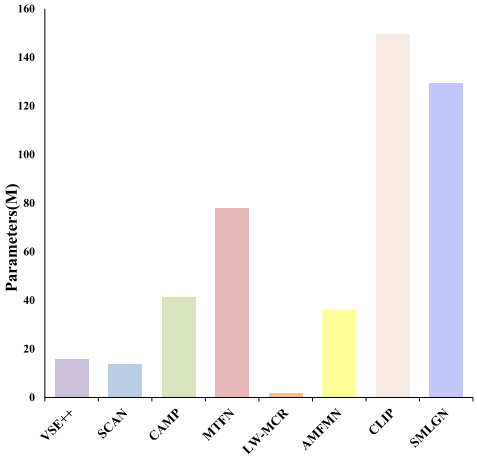


Fig. 11. Comparison of parameter quantities for different methods.

Further experiments and analyses about the effectiveness of these modules can be found in Section IV-F1.

To further evaluate the efficacy of our models, we proceed to compare all methods in terms of model complexity and runtime. Specifically, we initially computed the parameter quantity of each model as a representation of model complexity. The comparison results are depicted in Fig. 11. It is evident from the graph that due to the adoption of a dual transformer architecture, our model exhibits a certain degree of increase in parameter quantity compared to CNN-based models, with the exception of CLIP. Additionally, as LW-MCR is designed as a lightweight model, its parameter quantity is significantly lower than other methods. Furthermore, our method's parameter quantity stands moderately lower than that of CLIP and positions itself within the realm of Transformer-based models with moderate parameterization levels.

Moreover, for comparing runtime performance among these models, inference time and evaluation time were employed as evaluation metrics. Inference time refers to the duration required for completing one cross-modal similarity calculation while evaluation time indicates the time taken for computing similarities between all images and texts across different test sets. Multiple experiments were conducted and average values were utilized for final comparison purposes. Our comparative analysis results are presented in Table III. It can be observed from Table III that due to employing larger transformer models for feature extraction purposes, our method demonstrates longer inference and evaluation times compared to other approaches except CLIP. Particularly noteworthy is

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TABLE III
TIME COMPARISON OF DIFFERENT METHODS

Method	Evalution Time(s)			Inference Time(s)
	RSICD	RSITMD	UCM	
VSE++	56.93	11.41	4.34	0.0063
SCAN	85.87	16.22	6.18	0.1150
CAMP	60.60	13.05	5.20	0.0684
MTFN	78.33	13.54	4.71	0.0810
LW-MCR	28.29	6.79	2.51	0.0090
AMFMN	66.01	13.62	4.73	0.0764
CLIP	2864.50	591.04	205.26	3.3154
SMLGN	524.70	211.49	23.10	1.1863

TABLE IV
EXPERIMENTAL RESULTS OF PARAMETER EXPERIMENTS OF $\mathcal{L}_{\text{CEG}}$ ON THE RSITMD I2T DATASET

$\mathcal{L}_{CEG}$		I→T			T→I			mR
$\mathcal{L}_{CEG_v}$	$\mathcal{L}_{CEG_t}$	R@1	R@5	R@10	R@1	R@5	R@10	
0.1	0.9	15.04	37.17	51.99	14.20	40.71	56.77	35.98
0.2	0.8	15.71	38.50	49.78	**14.65**	42.83	59.07	36.76
0.3	0.7	14.16	35.18	48.45	11.64	41.06	57.04	34.59
0.4	0.6	14.38	35.62	50.66	12.70	39.91	56.37	34.94
0.5	0.5	17.26	39.38	51.55	13.19	**43.94**	**60.40**	**37.62**
0.6	0.4	14.82	36.28	51.99	12.57	42.39	59.96	36.33
0.7	0.3	**19.47**	38.27	50.00	13.63	42.43	59.42	37.21
0.8	0.2	17.92	**40.04**	**52.21**	13.85	40.49	58.85	37.23
0.9	0.1	15.71	35.62	50.66	13.76	42.61	57.61	36.00

TABLE V
EXPERIMENTAL RESULTS OF PARAMETER EXPERIMENTS OF $\mathcal{L}_{\text{MD}}$ ON THE RSITMD I2T DATASET

$\mathcal{L}_{MD}$		I→T			T→I			mR
$\mathcal{L}_{MD_v}$	$\mathcal{L}_{MD_t}$	R@1	R@5	R@10	R@1	R@5	R@10	
0.1	0.9	15.71	37.39	52.43	12.74	42.08	58.14	36.42
0.2	0.8	17.26	38.05	51.77	14.56	42.52	57.12	36.88
0.3	0.7	14.60	38.50	50.44	**13.50**	41.06	58.41	36.08
0.4	0.6	13.05	35.18	49.56	9.82	40.35	**62.04**	35.00
0.5	0.5	17.26	**39.38**	51.55	13.19	**43.94**	60.40	**37.62**
0.6	0.4	**17.92**	36.28	52.21	13.19	41.24	58.81	36.61
0.7	0.3	14.38	33.41	48.89	12.04	42.04	58.50	34.87
0.8	0.2	13.50	35.18	49.12	12.52	40.27	59.69	35.04
0.9	0.1	14.82	35.84	**52.21**	12.96	41.95	58.98	36.13

that LW-MCR showcases notably faster runtime owing to its lightweight design philosophy. However, when compared against CLIP which also utilizes Transformer architecture, our approach exhibits advantages in both inference and evaluation times positioning it as a moderately time-consuming model overall. Therefore, taking both aspects into consideration, despite a slight increase in the parameter quantity caused by our method, our inference and evaluation times remain within acceptable limits while achieving a significant enhancement in accuracy.

E. Parameters Analysis

In the multisubspace joint learning module, we construct the overall optimization objective of the SMLGN method as shown in (29). The hyperparameters μ and ν, respectively, control the contributions of the CEG and the GSMU. To study their effects on performance, we conduct extensive experiments on the RSITMD dataset and use grid search to adjust the parameters. We compare the R@1, R@5, and R@10 values of both parameters μ and ν for the I→V and V→I tasks. We also analyze the overall mR of these two tasks to assess the reasonability of the hyperparameters μ and ν. To observe the impact of different values of μ and ν on our method, we set the range of μ from 10 to 100 with an increment of 10, and the range of ν from 10 to 100 with an increment of 10. The setting of these ranges is based on a large number of experimental results to facilitate the intuitive observation of the optimal choice of hyperparameters. The parameter experimentation is conducted by changing one hyperparameter (e.g., ν) while keeping the other hyperparameter (e.g., μ) fixed. The specific implementation and analysis are as follows.

First, according to experience, we set ν to a fixed value of 30. Then, we increment the value of μ by 10 in each step, ranging from 10 to 100. We record the R@1/5/10 results for both the I→V and V→I tasks. As shown in Fig. 12, the mR value is the best when μ is set to 70. Although the $\mu = 80$ model performs best in terms of R@1, it shows average performance in R@5 and R@10. Therefore, considering various metrics, we choose $\mu = 70$ as the final value.

Next, with the value of μ fixed at 70, we conduct experiments on ν. We vary ν from 10 to 100, incrementing it by 10 in each step. Fig. 13 shows that when ν is set to 30, the model achieves the highest mR result and exhibits the best performance. Similarly, when ν is set to 80, although the precision of I→V R@1 is high, its performance in the other five metrics is inferior to that of $\nu = 30$. Therefore, we set $\nu = 30$.

In addition to the above, the trend graph of mR reveals some insights.

1) Regarding the weight of L_{CEG}, the experimental results show that since L_{CEG} and L_{MD} are a pair of constraint functions, increasing the weight of L_{CEG} can improve the degree of model antitraining, which in turn improves the overall retrieval performance. However, if the weight is too large, it may cause the model to undergo excessive adversarial learning. This can lead to an increased emphasis on modality discrimination and the failure to align the distributions of different modalities.
2) Regarding the weight of L_{GSMU}, the results suggest that reducing its weight appropriately can prevent the model from excessively focusing on intramodal sample discrimination while neglecting the crucial task of preserving the similarity of cross-modal samples.

Furthermore, as evident from (27) and (28), L_{CEG} and L_{MD} represent two sets of loss values ($L_{\text{CEG_}v}$and $L_{\text{CEG_}t}$, $L_{\text{MD_}v}$ and $L_{\text{MD_}t}$) derived from image features and text features, respectively, which are then combined using a weighted average approach. To validate the feasibility of this average weighting scheme (i.e., combining weights of 0.5 each), we conducted two sets of parameter experiments. As depicted in Tables IV and V, the corresponding variations in retrieval accuracy for the functions L_{CEG} and L_{MD} are presented under weight changes ranging from 0.1 to 0.9. The results indicate

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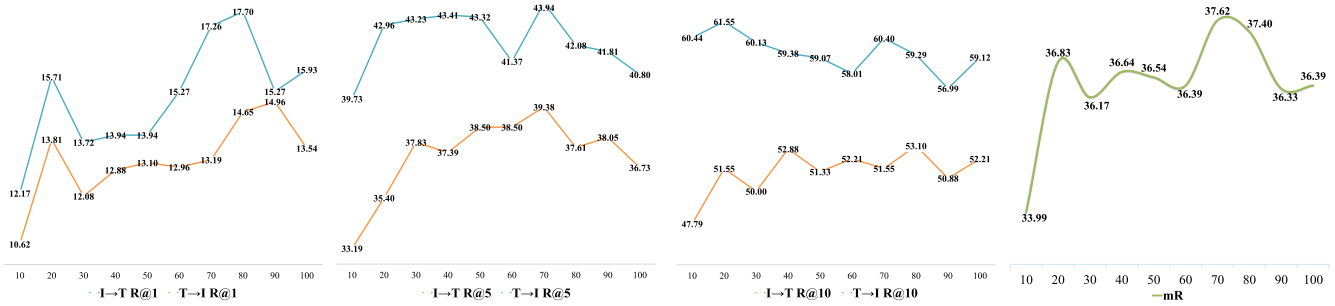


Fig. 12. Line plots of the experimental results for the parameter μ. They show the variations of R@k (k = 1, 5, 10) and mR for both the i2t and t2i tasks.

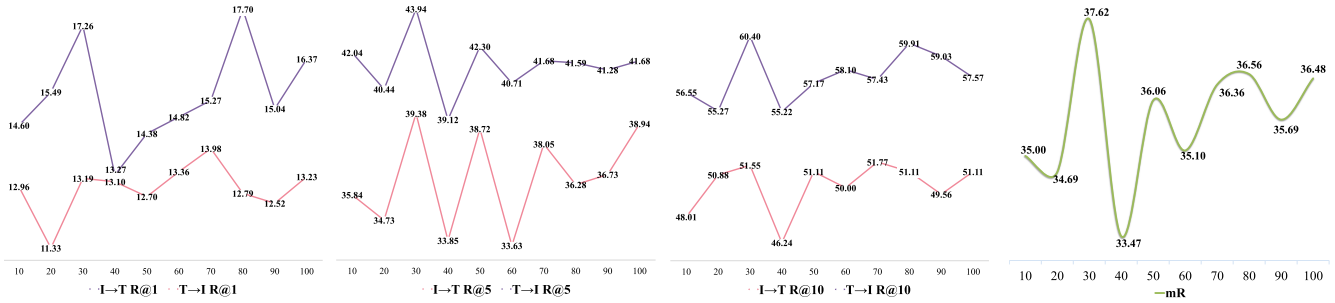


Fig. 13. Line plots of the experimental results for the parameter ν. They show the variations of R@k (k = 1, 5, 10) and mR for both the i2t and t2i tasks.

that although certain individual precision indices for either L_{CEG} or L_{MD} may be higher under alternative weight combinations than under average weighting, we prioritize overall accuracy considerations when opting for the average weighting scheme.

F. Ablation Study

To validate the effectiveness of our proposed SMLGN method and its individual components, we conducted extensive ablation experiments on the publicly available RS I2T dataset, RSITMD. These experiments included ablations on each module and on the distance functions in Section III-C.

1) Ablation Experiments on Individual Modules: In this section, we conducted a study on six variants of the proposed SMLGN method by investigating three key modules: 1) the importance of the GSMU; 2) the MLGJU composed of the ILMGU and the LEGRC; and 3) the role of the MAFL.

The implementation details of different variants are described as follows.

1) In the case of only having the GSMU, we introduce a variant (SMLGN w/o MLGJU&MAFL) of the proposed SMLGN method.
2) To validate the effectiveness of the GSMU, we set up the variant (SMLGN w/o GSMU) for discussion and verification.
3) With only the ILMGU and LEGRC retained, we perform a variant (SMLGN w/o GSMU&MAFL) of the SMLGN method.
4) We introduce the variant (SMLGN w/o MLGJU) to discuss the importance of the multilevel guidance mechanism.
5) A variant (SMLGN w/o GSMU&MLGJU) is created by keeping only the MAFL in the SMLGN.
6) By discarding the MAFL, we construct the variant (SMLGN w/o MAFL) to verify the effectiveness of adversarial learning and joint learning with multiple objective functions.

The experimental results are shown in Table VI. Based on the research findings, we conducted analyses from the following perspectives.

a) Global similarity metric unit (GSMU): The purpose of the GSMU is to fully leverage the feature information in the single-modal global subspace and constrain the overall feature representation across different modalities. This allows images or texts with high similarity to be closer in the global subspace, while those with larger differences are farther apart. By comparing the results of SMLGN w/o GSMU and SMLGN, it can be observed that the inclusion of GSMU improves the accuracy of mR. It also enhances the R@1 measure for both the I→V and V→I tasks, improving the precision of the model. Furthermore, we compare SMLGN w/o GSMU&MLGJU with SMLGN w/o MLGJU and find that the latter achieves significantly higher overall accuracy. These improvements indicate that the global similarity metric unit is effective in enhancing the discriminability of data within modalities and, consequently, improving retrieval accuracy.

b) Multilevel guided joint unit (MLGJU): The MLGJU consists of the combined effects of the ILMGU and the LEGRC. The multilevel guidance helps align the information between modalities and fuse local and global information, thereby bringing the distribution of different modal data closer in the common space and addressing the semantic gap issue. By comparing SMLGN with SMLGN w/o MLGJU, it can be observed that the MLGJU module improves the R@1 and R@5 measures for I→T by 1.33% and 4.20%, respectively, and for T→I by 0.54% and 2.08%, respectively.

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TABLE VI
EXPERIMENTAL RESULTS OF ABLATION EXPERIMENTS ON INDIVIDUAL MODULES ON THE RSITMD I2T DATASET

Ablation	GSMU	MLGJU	MAFL	I→T			T→I			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
SMLGN w/o MLGJU&MAFL	√			12.17	34.73	48.89	11.77	41.50	59.47	34.76
SMLGN w/o GSMU&MAFL		√		15.49	37.17	**52.21**	**14.65**	42.65	60.35	37.09
SMLGN w/o GSMU&MLGJU			√	15.93	34.73	51.55	14.29	42.35	60.31	36.53
SMLGN w/o MAFL	√	√		13.94	38.50	**52.21**	13.50	42.65	59.69	36.75
SMLGN w/o MLGJU	√		√	15.93	35.18	51.55	12.65	41.86	**62.26**	36.57
SMLGN w/o GSMU		√	√	15.49	38.50	**52.21**	12.43	43.58	60.31	37.09
SMLGN	√	√	√	**17.26**	**39.38**	51.55	13.19	**43.94**	60.40	**37.62**

Additionally, the overall mR is improved from 36.57% to 37.62%. Moreover, the SMLGN w/o MLGJU&MAFL method achieves higher precision in most R@k measures compared to the SMLGN w/o MAFL, with improvements of 41.97%, 13.36%, 5.43%, 12.04%, 5.89%, and 1.56% on the respective measures. The final mR measure is improved by 8.21%. The comparison results indicate that with the addition of the multilevel guided learning, the scores of various evaluation measures have improved. This is because the multilevel guided module aligns at a fine-grained level and enhances the global features with aligned local information. This results in a more effective representation of modal features and, hence, improves retrieval accuracy.

c) Modality adversarial fusion learning: The objective of MAFL is to discover a powerful common semantic space through adversarial learning. Its purpose is to reduce the discrepancy between representations of different modalities that share the same semantic meaning while simultaneously maximizing the dissimilarity between modalities with distinct semantics. According to the comparison between SMLGN w/o MAFL and SMLGN, it is observed that the inclusion of adversarial learning significantly improves the R@1 accuracy for I→T, but the R@1 accuracy for T→I fluctuates. However, the overall mR improvement from 36.75% to 37.63% serves as evidence of the effectiveness of MAFL. In addition, comparing SMLGN w/o GSMU&MAFL with SMLGN w/o GSMU, it is found that the removal of the MAFL module leads to a 1.33% (I→T) and 0.93% (T→I) increase in the R@5 measure. This indicates that the MAFL module has an impact on these two measures, while other measures and the overall mR remain unchanged. Combining this observation with the comparison to SMLGN, it is believed that this phenomenon occurs due to the lack of the constraint of intramodal metric loss in the GSMU module, leading to similar accuracy between the two. In conclusion, the inclusion of the deep adversarial fusion unit resolves the problem of distribution differences between modalities, promoting the even distribution of data in the common semantic space.

2) Ablation Experiment of Distance Metrics: In this section, we conduct experiments on the proposed SMLGN method using the four distance functions mentioned in Section III-C. The experimental results are shown in Table VII. The specific implementation details involve setting the distance metric functions used in the loss function L_{GSMU} of GSMU to cosine distance, MSD distance, $L1$ distance, and $L2$ distance, while keeping the rest unchanged. From Table VII, it can be observed that the cosine distance achieves the highest accuracy in terms of the mR metric. For the I→T and T→I tasks, the results of most R@k metrics also show that the cosine distance performs the best. We speculate that there are two possible reasons for this. First, cosine similarity distance primarily focuses on the directional relationship between text vectors and image vectors when calculating similarity rather than considering their absolute values. This enables it to effectively compare feature vectors across different modalities. Second, cosine similarity distance can overlook absolute changes and proportional differences between modalities, and instead, it emphasizes the directional relationship between vectors. This enables it to match and retrieve data with similar semantics or content across different modalities.

TABLE VII
EXPERIMENTAL RESULTS OF ABLATION EXPERIMENTS OF DIFFERENT DISTANCE METRICS ON THE RSITMD I2T DATASET

Ablation	I→T			T→I			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
SMLGN-L1	13.72	36.95	48.01	11.02	39.91	60.44	35.01
SMLGN-L2	13.72	36.06	50.66	12.30	41.42	**61.50**	35.94
SMLGN-msd	16.15	35.40	50.00	12.92	42.17	59.73	36.06
SMLGN-cosine	**17.26**	**39.38**	**51.55**	**13.19**	**43.94**	60.40	**37.62**

V. CONCLUSION

In this article, we propose the SMLGN method for RS I2T cross-modal retrieval. First, on the whole, combining local-global guidance within modalities and bidirectional guidance between modalities, this method can learn to obtain a common semantic space. The modal embeddings in the space are directly comparable and make up for the semantic differences between the modalities. Second, in terms of the significant variation within classes and minimal variation between classes in RS images, our method introduces the GSMU for clustering operation at the intramodal sample level, which helps to have interclass distinguishability and intraclass clustering within the modal data. Third, to address the critical issue of cross-modal tasks, we propose the MLGJU to

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guide intermodal local information alignment and intramodal global-local information fusion at the same time. Finally, the multisubspace joint learning module is introduced to optimize the model using a multiobjective loss function and adversarial joint learning on the obtained feature subspace. Experiments conducted on the public datasets RSICD, RSITMD, and UCM validate the effectiveness of our approach. In the future, we will explore how to extend the proposed SMLGN method to address large-scale cross-modal RS image retrieval, such as by introducing hash code learning into our work.

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